Nutritional intervention in patients with periodontal disease: clinical, immunological and microbiological variables during 12 months.

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The role of nutrition in onset, progression and treatment of periodontitis has not been thoroughly evaluated. In the present prospective clinical study, we investigated the influence of a nutritional intervention on changes in clinical, microbiological and immunological periodontal variables during a period of 12 months in patients with the metabolic syndrome and chronic periodontitis. Twenty female subjects with the metabolic syndrome and mild to moderate chronic periodontitis participated in a guided nutritional intervention programme. Examinations were assessed before, and at 2 weeks, 3, 6 and 12 months after intervention. Clinical measurements included probing depth, Löe and Silness gingival index and Quigley-Hein plaque index. In gingival crevicular fluid, periodontopathogens, levels of IL-1beta and IL-6 as well as the activity of granulocyte elastase were determined. In stimulated saliva, antioxidative and oxidative variables were measured. After 12 months the following significant changes could be observed: reduction of clinical probing depth (2.40 v. 2.20 mm; $P < 0.001$), reduction of gingival inflammation (gingival index 1.13 v. 0.9; $P < 0.001$), reduced concentrations of IL-1beta (4.63 v. 1.10 pg/ml per site; $P < 0.001$) as well as IL-6 (1.85 v. 0.34 pg/ml per site; $P = 0.022$) in gingival crevicular fluid. Bacterial counts in gingival crevicular fluid as well as oxidative and antioxidative variables in saliva showed no significant changes. Only salivary catalase showed a tendency to lower values. These findings indicate that in patients with the metabolic syndrome wholesome nutrition might reduce inflammatory variables of periodontal disease and promote periodontal health.